A Phase Vocoder Approach To Pitch Shifting

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1 Introduction

The manipulation of pitch in digital signals is not achieved by simply scaling frequencies, but through structured transformations in the time-frequency domain.

2 Background

2.1 Discrete Fourier Transform

The Discrete Fourier Transform is a fundamental tool used in digital signal processing that enables sophisticated analysis and synthesis of discrete-time signals [5]. The DFT transforms a signal from the time domain into its frequency-domain representation, allowing the signal to be expressed as a sum of sinusoids [5]. Mathematically, the DFT is defined as:

$$X[m] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi n \frac{m}{N}} \tag{1}$$

where N is the number of samples from the input signal $x[n]$, n denotes the time-domain index, m represents the frequency bin index, n is the index of the input sequenc in the time-domain and $j = \sqrt{-1}$. Each coeffecient in $X[m]$ is complex-valued and can be expressed in terms of magnitude and phase, which describe the contribution of a corresponding frequency component to the overall signal [6].

2.2 Windowing

Windowing is a function used to isolate parts of a discrete time signal so that analysis on the signal can be performed on contiguous blocks [1]. More specifically, the input signal is multiplied by a windowing function whose domain is nonzero for a particular range. This results in a small localized snapshot of the input signal, and every other value outside of the windows domain is reduced to zero [7]. Figure 1 below demonstrates the functionality of a window function.

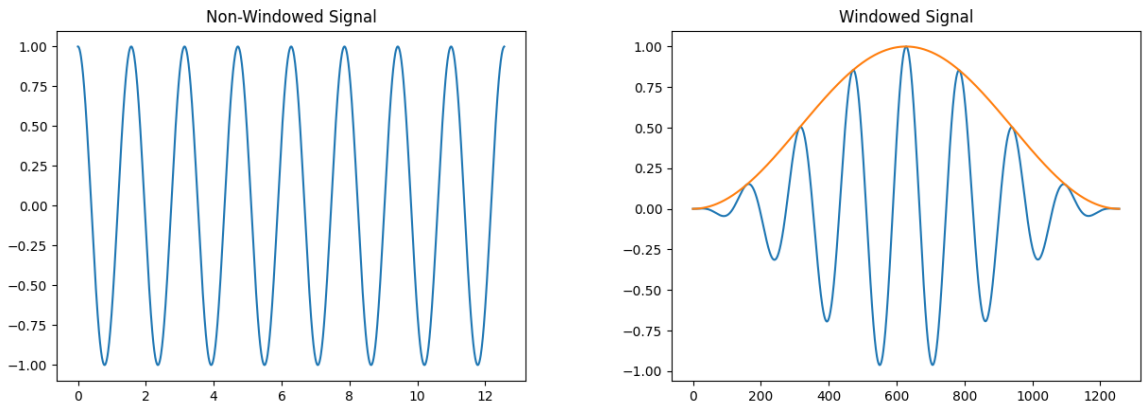

Figure 1: After applying a window function (highlighted by the orange cosine curve), the signal is tapered at the edges, resulting in a temporally localized signal.

As dipicted by Figure 1, after applying the window function to the example signal, the edges of the signal tend toward zero. In this example, the Hanning window function was used to shape the signal, however, there are multiple windowing functions such as the Rectangle window function, Triangle window function and Hamming window function [6]

2.3 Short-Time Fourier Transform

The Short-Time Fourier Transform (STFT) is a time-frequency representation used to compute the DFT of contiguous windowed segments of a signal. This is accomplished by multiplying the signal by a windowing function so that only a small portion of the signal is nonzero [2]. Mathematically, it is similar to equation (1) but with a few modifications: the addition of a windowing function, the domain of the input signal, and how the output is interpreted. Below is the STFT:

$$X[m,k] = \sum_{n=0}^{N-1} x[n+kH]w[n]e^{-2\pi jmn/N} \tag{2}$$

Where $X[m,k]$ is the 2-dimensional output array indexed by both frequency bin and time frame and $x[n+kH]$ is the section of the input signal determined by frame index and hop size [7]. The outcome of the STFT is a representation that transforms a signal from the time domain to the frequency domain. In this form, the signal's frequency components can be systematically modified and recombined to reconstruct a new signal.

3 Related Works

Pitch-shifting has been approached by a number of time–frequency methods. Early methods, such as Dolson's (1986), employed the use of the Short-Time Fourier Transform in conjunction with the phase vocoder algorithm. In particular, Dolson describes a pitch-shifting method

that relies on time-scale manipulation to modify the pitch of a given signal [2]. For example, by stretching a signal, making it longer in the time domain, and then rescaling it back to its original length, the overall frequency content of the signal is effectively decreased, which corresponds to lowering the signal's pitch. Dolson acknowledges, however, that this approach is imperfect: specific constraints must be enforced when performing a pitch shift, as failing to do so may introduce unwanted artifacts into the signal or result in a reconstructed signal that has lost information [2].

Another limitation of the standard phase-vocoder technique, as discussed in [4], is that only a single linear pitch-scale modification can be applied during processing. That is, all frequency components are multiplied by the same pitch-shifting factor. Consequently, generating harmonized outputs requires repeated application of the full algorithm [4]. This significantly increases computational cost and limits real-time feasibility.

These challenges lead to further innovations in the phase vocoder algorithm. Modern implementations shift away from the standard two-step timescale and resample approach and adopt an analysis and resynthesis paradigm [3]. The new approach leverages phase information obtained from the STFT step to more accurately reconstruct the modified signal[3].

4 Methodology

Dijkstra's algorithm was chosen for single-source shortest path computation on graphs with non-negative edge weights, following the implementation presented in CLRS [**?**].

5 Results

An exemplary figure is shown in Figure **??**. Any time a figure is used, it must have a capation and be directly referred to and discussed in the text.

6 Conclusion

References

[1] Barry, D., Dorran, D., and Coyle, E. Time and pitch scale modification: A real-time framework and tutorial.

[2] Dolson, M. The phase vocoder: A tutorial. *Computer Music Journal 10*, 4 (1986), 14–27.

[3] Götzen, A. D., Bernardini, N., and Arfib, D. Traditional (?) implementations of a phase-vocoder: The tricks of the trade. In *Proceedings of the COST G-6 Conference on Digital Audio Effects (DAFx-00)* (Verona, Italy, Dec. 2000), pp. 37–44.

[4] Laroche, J., and Dolson, M. New phase-vocoder techniques for pitch-shifting, harmonizing and other exotic effects. In *Proceedings of the IEEE Workshop on*

Applications of Signal Processing to Audio and Acoustics (WASPAA) (New Paltz, NY, USA, Oct. 1999), IEEE, pp. 91–94.

[5] Lyons, R. G. *Understanding Digital Signal Processing*, 3rd ed. Pearson, Upper Saddle River, NJ, USA, 2010.

[6] McFee, B. Digital signals theory, 2025.

[7] Royer, T. Pitch-shifting algorithm design and applications in music. Master's thesis, KTH Royal Institute of Technology, School of Electrical Engineering and Computer Science, 2019.